

INTERNSHIP REPORT PRESENTATION

Graphics Design and Product Prototyping for a Consumer Toy Brand

NITIN KUMAR CHOUDHARY

Roll No: 2410030504

B.Tech Computer Science & Engineering | Section 3CSE26 | Batch 2026-27

School of Computer Science & Engineering

IILM University, Greater Noida, U.P.

1

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Candidate & Internship Overview

A snapshot of the student profile and the internship engagement

1

Candidate Details

STUDENT NAME

Nitin Kumar Choudhary

ROLL NUMBER

2410030504

DEGREE

B.Tech Computer Science & Engineering

BATCH / SECTION

2026-27 | 3CSE26

INSTITUTION

IILM University, Greater Noida

2

Internship Profile

ORGANIZATION

Intelligentia Labs Pvt. Ltd.

BRAND

InventoJoy

ROLE

Graphics and Prototyping Intern

MODE

Work from office, Greater Noida

DURATION

26 March 2026 – [end date]

Project Overview

The project covers the complete printed system for a toy product — moving a design from brief through structural definition, graphic application, physical verification and manufacturing review to production-ready artwork released to a vendor.

1

Brief Intake

Objective, product, audience, channel,
deadline and approval path

2

Structural Design

Dieline construction: cuts, creases,
bleed, safety zones, glue flaps

3

Compliance Content

Age grading, warnings, origin and
barcode placed before layout

4

Graphic Application

Panel-by-panel hierarchy, illustration,
typography and colour

5

Prototyping

Paper folds, plotter-cut mock-ups,
substrate comps, 3D prints

6

DFM Review

Constraint check with the product and
manufacturing teams

7

Assembly Guides

Language-independent instruction
design, validated by observation

8

Pre-press & Handover

Checklist verification, versioned release
to the print vendor

Organization & Internship Profile

Details as stated in the appointment letter issued by the organisation



Intelligentia Labs Pvt. Ltd. — InventoJoy

Host organisation for the internship engagement · E-003, EPIP Site-5, UPSIDA, Greater Noida – 201306

1

DEPARTMENT

Design — Packaging
& Product

2

ROLE

Graphics and
Prototyping Intern

3

MODE

Work from office
10:00 – 19:30

4

DURATION

3 months minimum
from 26 March 2026

Note: Only information stated in the appointment letter is presented here. Company history, size, clients and revenue are not claimed.

Problem Statement & Industry Context

A packaging design is four different artefacts at once, each with its own constraint set



The Challenges

- A graphic approved on screen can break once the board is folded
- A pack that presents well on shelf can be uneconomic to ship
- Statutory content placed last ends up illegible or omitted
- Manufacturing limits surface after tooling is already committed
- Dieline errors are only discovered once the die has been cut
- Assembly instructions are ambiguous to everyone but their author
- An unapproved file reaching a vendor causes a production loss
- One-off files force the same work to be redone on the next product



The Proposed Approach

A gated, constraint-first design workflow in which structural, regulatory and manufacturing limits are admitted as parameters before creative work, and every decision is verified physically at the lowest cost that answers the question.

Constraint-first sequencing

Physical verification

Gated cross-functional review

Versioned vendor release

Core Project Objectives

Eight objectives derived from the responsibilities set out in the appointment letter

1

Apply graphic design fundamentals to toy packaging and printed collateral

2

Construct and validate dielines for boxes, inserts and stickers

3

Optimise packaging for manufacturing, shipping and shelf display

4

Build physical prototypes and support product testing

5

Design intuitive, child-friendly instruction and assembly guides

6

Ensure design-for-manufacturing readiness with the product team

7

Support marketing creatives for digital and print platforms

8

Apply version control and structured vendor handover discipline

Technologies & Toolchain

The tools applied across structural design, graphics, prototyping and release

Ai

Illustrator

Dielines, vector artwork, production files

Ps

Photoshop

Image preparation and mock-up compositing

Id

InDesign

Assembly guides, catalogues, trade sheets

Fg

Figma

Collaborative layout and review

F3

Fusion 360

Parametric part modelling and fit studies

3D

3D Printing

FDM and resin prototypes of components

Pl

Cutting Plotter

Accurate board mock-ups from dielines

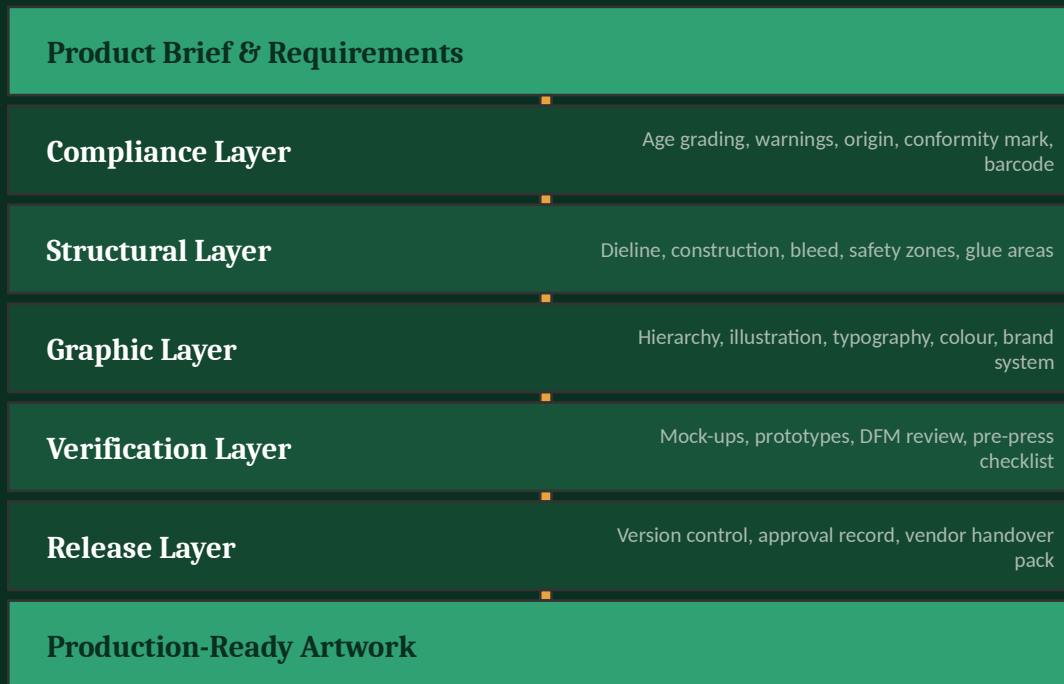
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Pre-flight

Separation, bleed and overprint verification

System Architecture

The design system as a constraint stack — each layer bounds the one above it



Project Methodology

Nine phases governing the execution of every deliverable

1

PHASE 1

Brief Intake

Parameters confirmed in writing

2

PHASE 2

Research & Reference

Category conventions and compliance

3

PHASE 3

Concept Development

Divergent options at low fidelity

4

PHASE 4

Structural Definition

Dieline fixed and physically folded

5

PHASE 5

Artwork Development

Graphics within the fixed template

6

PHASE 6

Mock-up & Iteration

Print, cut, fold, assess, repeat

7

PHASE 7

DFM Review

Joint constraint check as a gate

8

PHASE 8

Revision & Pre-press

Checklist executed before release

9

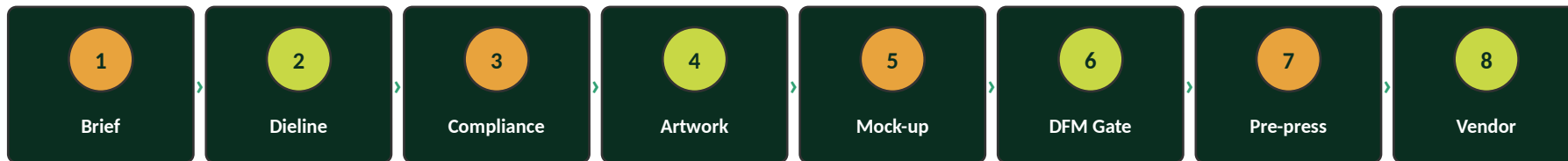
PHASE 9

Handover & Archive

Versioned release and approval record

End-to-End Design Workflow

How a single packaging artefact moves from brief to production



Stage Descriptions

1. Brief

Objective, product, channel, deadline and approval path fixed in writing.

2. Dieline

Structure selected, constructed, and physically folded before artwork.

3. Compliance

Statutory content placed on the dieline before creative layout.

4. Artwork

Hierarchy resolved panel by panel within the safety zones.

5. Mock-up

Paper fold, then board, then substrate comp — cheapest first.

6. DFM Gate

Constraints satisfied or formally accepted as deviations.

7. Pre-press

Fixed checklist: separations, bleed, links, barcode, statutory text.

8. Vendor

PRODUCTION-status file released with a written approval record.

Expected Project Outcomes

Anticipated results of the approach — not measured or verified achievements



Production-ready packaging artwork released to a print vendor



Validated dielines for boxes, inserts and sticker sheets



Physical prototypes verifying fit, containment and durability



A child-friendly assembly guide validated by observation



Reduced late-stage revisions through early DFM consultation



A reusable template and dieline reference for the product range



Correct statutory content accommodated on every pack



A versioned archive guaranteeing reprintable production files

No quantitative measurement was instituted during the internship; these are anticipated outcomes of the approach, stated as such.

Learning & Technical Exposure

Concepts and skills engaged with during the internship



Structural packaging design



Colour management for print



Additive prototyping (3D printing)



Technical & instructional illustration



Toy labelling and conformity context



Print production & pre-press



Substrate and finish selection



Physical model making



Design for manufacturing



Version control & vendor handover

Internship Certificate

[INSERT SCANNED INTERNSHIP CERTIFICATE HERE]

The certificate must be issued and signed by Intelligentia Labs Pvt. Ltd. Request it from your reporting manager, then insert the scan on this slide.

Engagement Details

ORGANIZATION

Intelligentia Labs Pvt. Ltd.

BRAND

InventoJoy

ROLE

Graphics and Prototyping Intern

MODE

Work from office

DURATION

26 March 2026 - [end date]



Thank You

Questions welcome

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